Submission 1: PTX3 Expression, Vitamin D and Preeclampsia (GSE85307)

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Variables of Interest

- **Chosen Gene:** *PTX3* (pentraxin 3), an inflammatory marker previously linked to preeclampsia (Reference: 2)
- **Continuous covariate:** maternal baseline 25-hydroxyvitamin D (ng/mL) (mat_vitID)
- **Categorical covariate 1:** pregnancy outcome (Normal vs Preeclampsia) (pre_preeclampsia)
- **Categorical covariate 2 (derived from a continuous variable):** vitamin D status from `mat_vitD` (vitD_status) (Deficient $<$ 20, Insufficient 20–29.9, Sufficient 30 ng/mL)

```
# loading tidyverse library
library(tidyverse)
```

Load the raw files

```
covar <- read_csv("data/GSE85307_covar-1.csv")
exp <- read_csv("data/GSE85307_geneExpression.csv")
geneNames <- read_csv("data/GSE85307_geneNames.csv")

# Quick look at each file
glimpse(covar)
dim(exp) # 500 probes (rows) with 157 samples (+ 1 additional probe ID column)
head(geneNames)
```

Data Cleaning

Cleaning the covariate data

Each covariate value is stored as text such as `"bmi_kg/m2: 37.52"`, so keep only the part after `": "` and convert numbers to numeric.

```
covar_clean <- covar |>
  # Rename first column name by position
  rename(subjectID = 1) |>
  # For every column except the IDs, keep only the text after ": "; consulted
  # the help of Claude for this step to help with str_removal, detecting patterns
  mutate(across(-c(subjectID, geoID),
                ~ str_remove(.x, "^.*: "))) |>
  # Convert the continuous variables from text to numeric
  mutate(across(c(mat_age_yrs,
                  delivery_gestAge_wks,
                  sampleCollection_gestAge_wks,
```

```
                  mat_BMI,
                  mat_vitD),
                as.numeric)) |>
  mutate(
    # Pregnancy outcome as a factor, with normal and preeclampsia levels
    preg_preeclampsia = factor(preg_preeclampsia,
                               levels = c("Normal", "Preeclampsia")),
    # Turn asthma into a labeled factor, either 0 or 1
    mat_asthma = factor(mat_asthma,
                        levels = c("0", "1"),
                        labels = c("Healthy", "Asthma")),
    # convert to factor variable
    batch_Microarray = factor(batch_Microarray),
    # turn continuous mat_vitD to categorical vitD_status variable
    vitD_status = cut(mat_vitD,
                      breaks = c(-Inf, 20, 30, Inf),
                      right  = FALSE,   # intervals are [low, high)
                      labels = c("Deficient (<20)",
                                 "Insufficient (20-29.9)",
                                 "Sufficient (30+)"))
  )

summary(covar_clean)
count(covar_clean, preg_preeclampsia, vitD_status)
```

Attach gene names using left_join; matching gene names

```
exp_named <- exp |>
  # PROBE-ID column has no header, so rename it
  rename(PROBEID = 1) |>
  # add the gene column
  left_join(geneNames, by = "PROBEID") |>
  relocate(GENE, .after = PROBEID)

# Check that every probe matched a gene name and there are no duplicate genes
sum(is.na(exp_named$GENE))
any(duplicated(exp_named$GENE))

# Keep only PTX3, change from wide to long format
ptx3 <- exp_named |>
  filter(GENE == "PTX3") |>
  pivot_longer(cols = starts_with("GSM"),
               names_to  = "geoID",
               values_to = "PTX3_exp")

head(ptx3)
```

Combine covariate data with gene expression data using geoID

```
# Check that the sample IDs are the same set in both files before joining
setequal(ptx3$geoID, covar_clean$geoID)

analysis_df <- covar_clean |>
```

```
  inner_join(ptx3, by = "geoID")

# 157 rows: one per sample, none lost or duplicated
nrow(analysis_df)

n_distinct(analysis_df$geoID)

# Spot-check one sample against the raw expression file
analysis_df |>
  filter(geoID == "GSM2264346") |>
  pull(PTX3_exp)

exp |>
  rename(PROBEID = 1) |>
  filter(PROBEID == 8083594) |>
  pull(GSM2264346)
```

Submission 1 Plots

Shared Theme

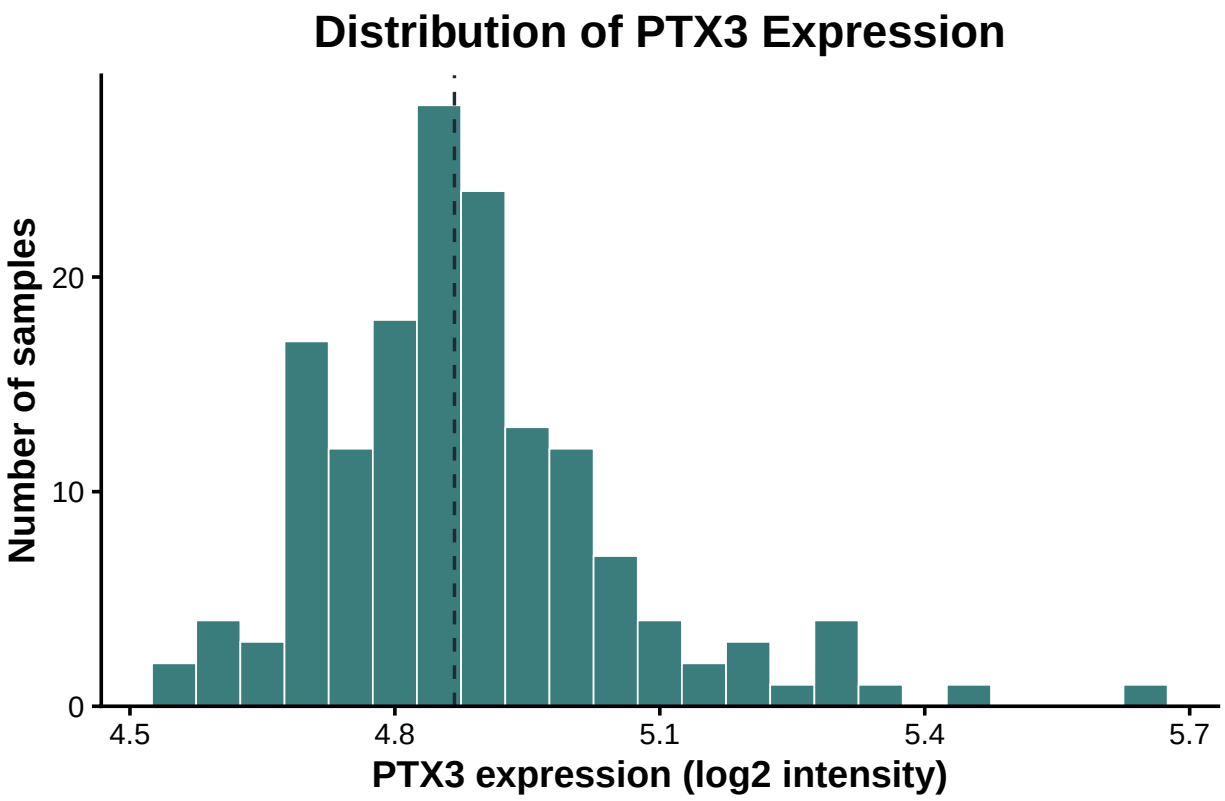
```
# One theme for consistency across all plots; publication quality
theme_pub <- theme_classic(base_size = 14) +
  theme(plot.title = element_text(face = "bold", hjust = 0.5),
        axis.title = element_text(face = "bold"),
        legend.position = "top",
        legend.title = element_text(face = "bold"))

# shared colors
outcome_colors <- c("Normal" = "#4C72B0", "Preeclampsia" = "#C44E52")
```

Histogram of PTX3 gene expression

```
hist <- ggplot(analysis_df, aes(x = PTX3_exp)) +
  # creating the histogram
  geom_histogram(binwidth = 0.05,
                 fill = "#3A7D7C",
                 color = "white",
                 linewidth = 0.3) +
  # median used instead of mean because robust to right tail
  # want to show where median is in the distribution
  geom_vline(aes(xintercept = median(PTX3_exp)),
             linetype = "dashed", color = "#1F2A37", linewidth = 0.6) +
  # leave some headroom above the tallest bar, 5%
  scale_y_continuous(expand = expansion(mult = c(0, 0.05))) +
  # Titles and axis labels
  labs(title = "Distribution of PTX3 Expression",
       x = "PTX3 expression (log2 intensity)",
       y = "Number of samples",
       caption = "Dashed line = median; n = 157") +
  # apply shared publication theme across all 3 figures
  theme_pub
```

```
hist
```

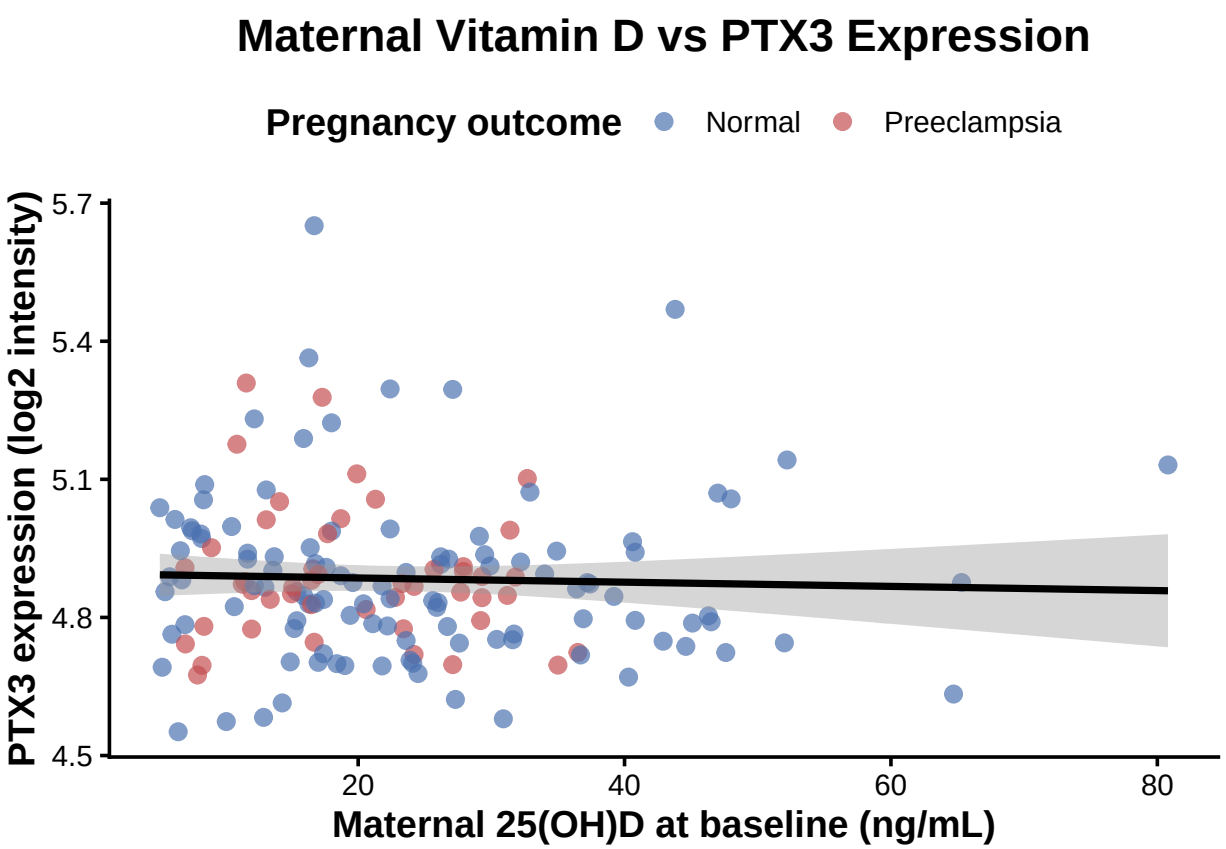



```
# save in figures folder so don't need to regenerate again
# ggsave("figures/histogram_PTX3.png",
#        hist,
#        width = 7,
#        height = 5,
#        dpi = 300)
```

Scatterplot comparing vitamin D vs. PTX3 gene expression

```
scatter <- ggplot(analysis_df, aes(x = mat_vitD, y = PTX3_exp)) +
  # plot each sample as a point, adjusting for transparency and size of point
  geom_point(aes(color = preg_preeclampsia),
             alpha = 0.7,
             size = 2.5) +
   # overall linear trend + 95% CI
  geom_smooth(method = "lm",
              color = "black",
              se = TRUE) +
  # so colors are same in every figure
  scale_color_manual(values = outcome_colors) +
  labs(title = "Maternal Vitamin D vs PTX3 Expression",
       x = "Maternal 25(OH)D at baseline (ng/mL)",
       y = "PTX3 expression (log2 intensity)",
       color = "Pregnancy outcome") +
  theme_pub
```

```
scatter
```



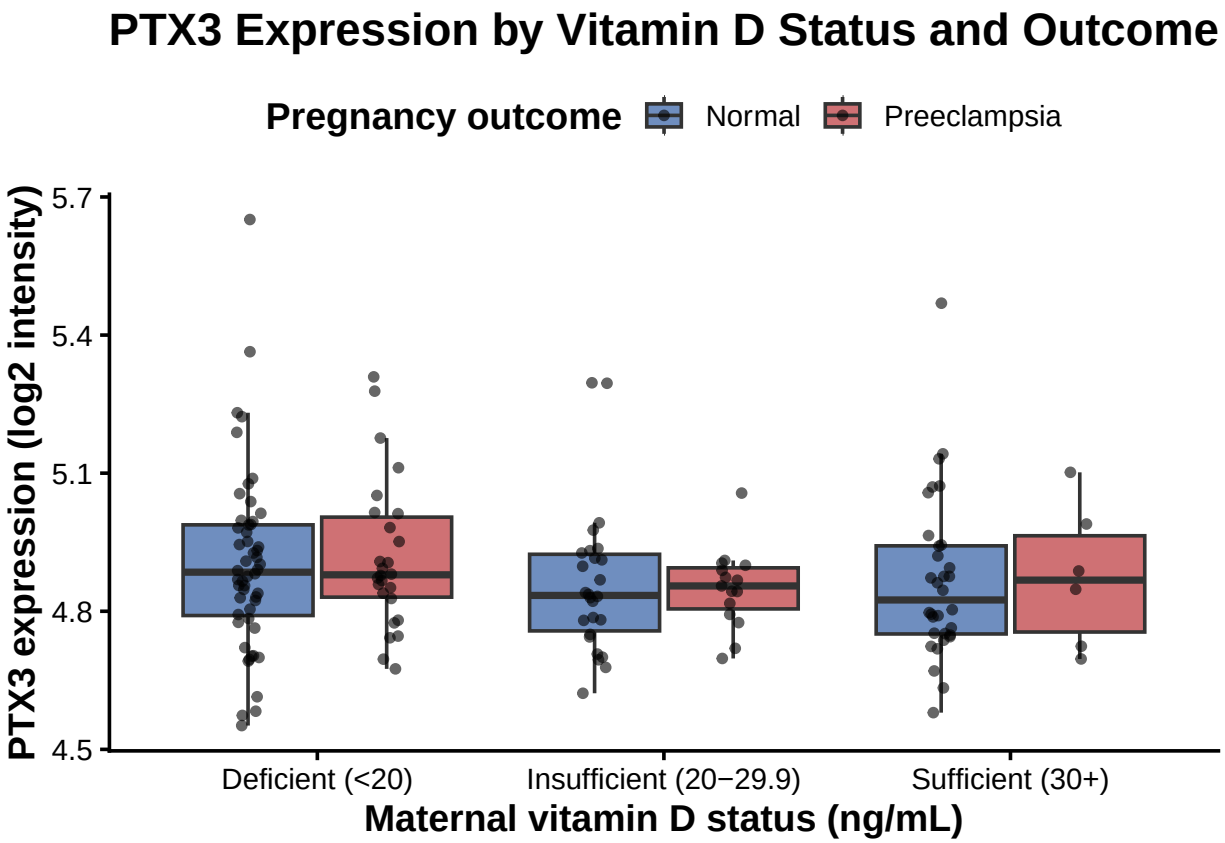

```
# save in figures folder so don't need to regenerate again
# ggsave("figures/scatter_vitD_PTX3.png",
#        scatter,
#        width = 7,
#        height = 5,
#        dpi = 300)
```

8. Boxplot of PTX3 by vitamin D status and pregnancy outcome

```
box <- ggplot(analysis_df, aes(x = vitD_status, y = PTX3_exp,
                               fill = preg_preeclampsia)) +
  # position_dodge makes Normal and Preeclampsia boxes side by side
  geom_boxplot(outlier.shape = NA,
               alpha = 0.8,
               position = position_dodge(width = 0.8)) +
  # Overlay individual samples so can see group sizes directly
  # position_jitterdodge so it spread points horizontally to reduce overlap
  # lines them up witoh their matching box using dodge = 0.8
  geom_point(position = position_jitterdodge(jitter.width = 0.15,
                                             dodge.width = 0.8),
             size = 1.2,
             alpha = 0.6) +
  # same colors for consistency
  scale_fill_manual(values = outcome_colors) +
```

```
  labs(title = "PTX3 Expression by Vitamin D Status and Outcome",
       x = "Maternal vitamin D status (ng/mL)",
       y = "PTX3 expression (log2 intensity)",
       fill = "Pregnancy outcome") +
  theme_pub

box
```




```
# save in figures folder so don't need to regenerate again
# ggsave("figures/boxplot_PTX3_vitD_outcome.png",
#        box,
#        width = 7,
#        height = 5,
#        dpi = 300)
```

References

1. Mirzakhani H, Litonjua AA, McElrath TF, et al. Early pregnancy vitamin D status and risk of preeclampsia. *J Clin Invest.* 2016;126(12):4702–4715. doi:10.1172/JCI89031. (Data source: GEO accession GSE85307.)

2. Xiong Z, Wang X, Jiang S, Jin M, Chen W. Association between pentraxin-3 and the risk of preeclampsia: a meta-analysis. *Medicine (Baltimore).* 2020;99(26):e20744. doi:10.1097/MD.0000000000020744

3. Anthropic. Claude (Opus 5.5) [Large language model]. 2026. https://claude.ai. Accessed October 25–26, 2026.

AI use statement: Claude (Anthropic) was used to help with str_removal for initial data cleaning and some GitHub debugging issues at initial setup. Also consulted Claude for debugging help during data cleaning.

Searched up different geom_ function via Google for help making the figures look nicer.